

SOLIDWORKS API (.NET VBA C#) — Reference (14/337): IAssemblyDoc (part 2 of 5)

SOLIDWORKS API Documentation (converted from CHM)

IAssemblyDoc (part 2 of 5) (Type)

SOLIDWORKS API Help

[Send comments](#) on this topic.

CompConfigProperties4 Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : CompConfigProperties4 Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::CompConfigProperties5](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CompConfigProperties4( _  
    ByVal Suppression As System.Integer, _  
    ByVal Solving As System.Integer, _  
    ByVal Visibility As System.Boolean, _  
    ByVal UseNamedRefConfig As System.Boolean, _  
    ByVal RefConfigName As System.String, _  
    ByVal ExcludeFromBOM As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim Suppression As System.Integer  
Dim Solving As System.Integer  
Dim Visibility As System.Boolean  
Dim UseNamedRefConfig As System.Boolean  
Dim RefConfigName As System.String  
Dim ExcludeFromBOM As System.Boolean  
Dim value As System.Boolean
```

```
value = instance.CompConfigProperties4(Suppression, Solving, Visibility, UseNamedRefConfig,  
RefConfigName, ExcludeFromBOM)
```

C#

```
System.bool CompConfigProperties4(  
    System.int Suppression,  
    System.int Solving,  
    System.bool Visibility,  
    System.bool UseNamedRefConfig,  
    System.string RefConfigName,  
    System.bool ExcludeFromBOM  
)
```

C++/CLI

```
System.bool CompConfigProperties4(  
& System.int Suppression,  
& System.int Solving,  
& System.bool Visibility,  
& System.bool UseNamedRefConfig,  
& System.String^ RefConfigName,  
& System.bool ExcludeFromBOM  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Suppression

Suppression state of this component as defined in [swComponentSuppressionState_e](#)

Solving

Solving state of this component as defined in [swComponentSolvingOption_e](#)

Visibility

True if you want to show the selected component in the graphics display area, false if not

UseNamedRefConfig

Not used

RefConfigName

- If a non-empty string is specified, then the referenced configuration of the selected component is changed to this named configuration

- If an empty string is specified, then the default referenced configuration is used

ExcludeFromBOM
True to exclude the configuration from the BOM, false to not

Return Value

True if setting the properties of the selected component in the specified configuration is successful, false if not

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::CompConfigProperties4](#).

Remarks

You can use configurations to save certain display characteristics with each of the assembly components and retrieve that configuration in the future. SOLIDWORKS Design applies the settings that you specify with this method to the active configuration.

You cannot set a component to lightweight.

Known reasons for failure of this method include:

- Invalid suppression state specified.
- Not preselecting a component.

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13

SOLIDWORKS API Help

[Send comments on this topic.](#)

CompConfigProperties5 Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : CompConfigProperties5 Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::CompConfigProperties6](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CompConfigProperties5( _  
    ByVal Suppression As System.Integer, _  
    ByVal Solving As System.Integer, _  
    ByVal Visibility As System.Boolean, _  
    ByVal UseNamedRefConfig As System.Boolean, _  
    ByVal RefConfigName As System.String, _  
    ByVal ExcludeFromBOM As System.Boolean, _  
    ByVal IsEnvelope As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim Suppression As System.Integer  
Dim Solving As System.Integer  
Dim Visibility As System.Boolean  
Dim UseNamedRefConfig As System.Boolean  
Dim RefConfigName As System.String  
Dim ExcludeFromBOM As System.Boolean  
Dim IsEnvelope As System.Boolean  
Dim value As System.Boolean
```

```
value = instance.CompConfigProperties5(Suppression, Solving, Visibility, UseNamedRefConfig,  
RefConfigName, ExcludeFromBOM, IsEnvelope)
```

C#

```

System.bool CompConfigProperties5(
    System.int Suppression,
    System.int Solving,
    System.bool Visibility,
    System.bool UseNamedRefConfig,
    System.string RefConfigName,
    System.bool ExcludeFromBOM,
    System.bool IsEnvelope
)

```

C++/CLI

```

System.bool CompConfigProperties5(
    & System.int Suppression,
    & System.int Solving,
    & System.bool Visibility,
    & System.bool UseNamedRefConfig,
    & System.String^ RefConfigName,
    & System.bool ExcludeFromBOM,
    & System.bool IsEnvelope
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Suppression

Suppression state of this component as defined in [swComponentSuppressionState_e](#)

Solving

Solving state of this component as defined in [swComponentSolvingOption_e](#)

Visibility

True if you want to show the selected component in the graphics area, false if not

UseNamedRefConfig

Not used

RefConfigName

- If a non-empty string is specified, then the referenced configuration of the selected component is changed to this named configuration
- If an empty string is specified, then the default referenced configuration is used

ExcludeFromBOM

True to exclude the configuration from the BOM, false to not

IsEnvelope

True if the component is an envelope, false if not

Return Value

True if setting the properties of the selected component in the specified configuration is successful, false if not

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::CompConfigProperties5](#).

Remarks

You can use configurations to save certain display characteristics with each of the assembly components and retrieve that configuration in the future. SOLIDWORKS Design applies the settings that you specify with this method to the active configuration.

You cannot set a component to lightweight.

Known reasons for failure of this method include:

- Invalid suppression state specified.
- Not preselecting a component.

See Also

[IAssemblyDoc Interface](#)

[IAssemblyDoc Members](#)

[IComponent2::SetSuppression2 Method \(\)](#)

[IComponent2::ReferencedConfiguration Property \(\)](#)

[IComponent2::IsEnvelope Method \(\)](#)

[IComponent2::Solving Property \(\)](#)

[IComponent2::ExcludeFromBOM Property \(\)](#)

[IComponent2::Visible Property \(\)](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

CompConfigProperties6 Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : CompConfigProperties6 Method (IAssemblyDoc)

Sets the properties for the selected [component](#) in the specified [configuration](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CompConfigProperties6( _  
    ByVal Suppression As System.Integer, _  
    ByVal Solving As System.Integer, _  
    ByVal Visibility As System.Boolean, _  
    ByVal UseNamedRefConfig As System.Boolean, _  
    ByVal RefConfigName As System.String, _  
    ByVal ExcludeFromBOM As System.Boolean, _  
    ByVal IsEnvelope As System.Boolean, _  
    ByVal SaveAssemblyAsPartOption As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim Suppression As System.Integer  
Dim Solving As System.Integer  
Dim Visibility As System.Boolean  
Dim UseNamedRefConfig As System.Boolean  
Dim RefConfigName As System.String  
Dim ExcludeFromBOM As System.Boolean  
Dim IsEnvelope As System.Boolean  
Dim SaveAssemblyAsPartOption As System.Integer  
Dim value As System.Boolean
```

```
value = instance.CompConfigProperties6(Suppression, Solving, Visibility, UseNamedRefConfig,  
RefConfigName, ExcludeFromBOM, IsEnvelope, SaveAssemblyAsPartOption)
```

C#

```
System.bool CompConfigProperties6(  
    System.int Suppression,  
    System.int Solving,  
    System.bool Visibility,  
    System.bool UseNamedRefConfig,  
    System.string RefConfigName,  
    System.bool ExcludeFromBOM,  
    System.bool IsEnvelope,  
    System.int SaveAssemblyAsPartOption  
)
```

C++/CLI

```
System.bool CompConfigProperties6(  
& System.int Suppression,  
& System.int Solving,  
& System.bool Visibility,  
& System.bool UseNamedRefConfig,  
& System.String^ RefConfigName,  
& System.bool ExcludeFromBOM,  
& System.bool IsEnvelope,  
& System.int SaveAssemblyAsPartOption  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Suppression

Suppression state of the selected component as defined in [swComponentSuppressionState_e](#)

Solving

Solving state of the selected component as defined in [swComponentSolvingOption_e](#)

Visibility

True if you want to show the selected component in the graphics area, false if not

UseNamedRefConfig

Not used

RefConfigName

- If a non-empty string is specified, then the referenced configuration of the selected component is changed to this named configuration
- If an empty string is specified, then the default referenced configuration is used

ExcludeFromBOM

True to exclude the configuration from the BOM, false to not

IsEnvelope

True if the selected component is an envelope, false if not

SaveAssemblyAsPartOption

Option for the selected component when saving this assembly as a part as defined in [swASMSLDPRTCompPref_e](#)

Return Value

True if setting the properties of the selected component in the specified configuration is successful, false if not

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::CompConfigProperties6](#).

Example

[Change Component to Envelope \(VBA\)](#)

[Change Component to Envelope \(VB.NET\)](#)

[Change Component to Envelope \(C#\)](#)

Remarks

You can use configurations to save certain display characteristics with each of the assembly components and retrieve that configuration in the future. SOLIDWORKS Design applies the settings that you specify with this method to the active configuration.

You cannot set the selected component to lightweight using this method.

Known reasons for failure of this method include:

- Invalid suppression state specified in Suppression.
- Not pre-selecting the component.

See Also

[IAssemblyDoc Interface](#)

[IAssemblyDoc Members](#)

[IComponent2::ExcludeFromBOM Property \(\)](#)

[IComponent2::Solving Property \(\)](#)

[IComponent2::Visible Property \(\)](#)

[IComponent2::ReferencedConfiguration Property \(\)](#)

[IComponent2::IsEnvelope Method \(\)](#)

[IComponent2::SetSuppression2 Method \(\)](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ComponentReload Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : ComponentReload Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::ComponentReload2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub ComponentReload()
```

Visual Basic (Usage)

```
Dim instance As IAssemblyDoc
```

```
instance.ComponentReload()
```

C#



```
void ComponentReload()
```

C++/CLI

```
void ComponentReload();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::ComponentReload](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

ComponentReload2 Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : ComponentReload2 Method (IAssemblyDoc)

Reloads and/or sets the read-only state of the specified component.

.NET Syntax

Visual Basic (Declaration)

```
Function ComponentReload2( _  
    ByVal Component As System.Object, _  
    ByVal ReadOnly As System.Boolean, _  
    ByVal Options As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim Component As System.Object  
Dim ReadOnly As System.Boolean  
Dim Options As System.Integer  
Dim value As System.Integer
```

```
value = instance.ComponentReload2(Component, ReadOnly, Options)
```

C#

```
System.int ComponentReload2(  
    System.object Component,  
    System.bool ReadOnly,  
    System.int Options  
)
```

C++/CLI

```
System.int ComponentReload2(  
& System.Object^ Component,  
& System.bool ReadOnly,  
& System.int Options  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Component
[IComponent2](#)

ReadOnly

True to set Component read-only after reload, false to allow write access

Options

Reload option as defined by [swComponentReloadOption_e](#)

Return Value

Error code as defined by [swComponentReloadError_e](#)

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::ComponentReload2](#).

Example

'VBA

'Preconditions:

'1. Open *public_documents\SOLIDWORKS\SOLIDWORKS 2021\samples\tutorial\apilarm2.sldasm*.

'2. Open the Immediate window.

'Postconditions:

' Inspect the Immediate window.

'=====

```
Dim swApp As SldWorks.SldWorks
Dim SWMODEL As SldWorks.ModelDoc2
Dim SWASSY As SldWorks.AssemblyDoc
Dim swcomponent As SldWorks.Component2
Dim Error As swComponentReloadError_e
Option Explicit
Sub main()
    Set swApp = Application.SldWorks
    Set SWMODEL = swApp.ActiveDoc
    Set SWASSY = SWMODEL
    Set swcomponent = SWASSY.GetComponentByName("secondgrip-1")
    Debug.Print swcomponent.GetPathName
    Error = SWASSY.ComponentReload2(swcomponent, True, swComponentReloadOption_e.swDontReloadOldComponents)
    Debug.Print "Error code: " & Error
End Sub
```

Remarks

This method is analogous to the Reload dialog that appears when you right-click on an assembly component in the FeatureManager design tree and select **Reload**. For more information, read the **SOLIDWORKS Design user-interface Help > Fundamentals > Document Basics > Multi-User Environment > Reload** topic.

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::ReplaceComponents2 Method \(\)](#)
[IModelDoc2::ReloadOrReplace Method \(\)](#)

Availability

SOLIDWORKS 2001Plus

SOLIDWORKS API Help

[Send comments on this topic.](#)

CopyWithMates Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : CopyWithMates Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::CopyWithMates2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CopyWithMates( _  
    ByVal ComponentsToCopy As System.Object, _  
    ByVal Repeat As System.Object, _  
    ByVal NewEntityToMateTo As System.Object, _  
    ByVal Values As System.Object, _
```

```
ByVal FlipAlignment As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim ComponentsToCopy As System.Object  
Dim Repeat As System.Object  
Dim NewEntityToMateTo As System.Object  
Dim Values As System.Object  
Dim FlipAlignment As System.Object  
Dim value As System.Boolean
```

```
value = instance.CopyWithMates(ComponentsToCopy, Repeat, NewEntityToMateTo, Values, FlipAlignment)
```

C#

```
System.bool CopyWithMates(  
    System.object ComponentsToCopy,  
    System.object Repeat,  
    System.object NewEntityToMateTo,  
    System.object Values,  
    System.object FlipAlignment  
)
```

C++/CLI

```
System.bool CopyWithMates(  
& System.Object^ ComponentsToCopy,  
& System.Object^ Repeat,  
& System.Object^ NewEntityToMateTo,  
& System.Object^ Values,  
& System.Object^ FlipAlignment  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ComponentsToCopy
Array of [components](#) to copy

Repeat
Array of boolean values; each value indicates whether to use the existing mate reference for the corresponding component to copy: if true, copies the existing mate reference; if false, uses the corresponding entry in the NewEntityToMateTo array for the new mate reference

NewEntityToMateTo
Array of new mate references that map to the Repeat arrays; if an entry in the Repeat array is false, then the corresponding entry in this array is the new entity with which to mate the component to copy

Values
Array of distance or angle values for the mate references; specify distance in meters and angle in radians; valid for for distance and angle mates only

FlipAlignment
Array of booleans that map to the NewEntityToMateTo array; each value indicates the corresponding mate's alignment; true to flip alignment, false otherwise; valid for distance and angle mates only

Return Value

True if calling this method succeeded; false if it failed

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::CopyWithMates](#).

Example

[Copy Components With Mates to Assembly \(VBA\)](#)
[Copy Components With Mates to Assembly \(VB.NET\)](#)
[Copy Components With Mates to Assembly \(C#\)](#)

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

Availability

CopyWithMates2 Method (IAssemblyDoc)

[See Also](#) [Example](#)[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : CopyWithMates2 Method (IAssemblyDoc)

Copies one or more components with mates in this assembly.

 .NET Syntax

Visual Basic (Declaration)

```

Function CopyWithMates2( _
    ByVal ComponentsToCopy As System.Object, _
    ByVal Repeat As System.Object, _
    ByVal NewEntityToMateTo As System.Object, _
    ByVal Values As System.Object, _
    ByVal FlipAlignment As System.Object, _
    ByVal FlipDimension As System.Object, _
    ByVal LockRotation As System.Object, _
    ByVal Orientation As System.Object _
) As System.Boolean

```

Visual Basic (Usage)



```

Dim instance As IAssemblyDoc
Dim ComponentsToCopy As System.Object
Dim Repeat As System.Object
Dim NewEntityToMateTo As System.Object
Dim Values As System.Object
Dim FlipAlignment As System.Object
Dim FlipDimension As System.Object
Dim LockRotation As System.Object
Dim Orientation As System.Object
Dim value As System.Boolean

```

```

value = instance.CopyWithMates2(ComponentsToCopy, Repeat, NewEntityToMateTo, Values, FlipAlignment,
FlipDimension, LockRotation, Orientation)

```

C#

```

System.bool CopyWithMates2(
    System.object ComponentsToCopy,
    System.object Repeat,
    System.object NewEntityToMateTo,
    System.object Values,
    System.object FlipAlignment,
    System.object FlipDimension,
    System.object LockRotation,
    System.object Orientation
)

```

C++/CLI

```

System.bool CopyWithMates2(
    & System.Object^ ComponentsToCopy,
    & System.Object^ Repeat,
    & System.Object^ NewEntityToMateTo,
    & System.Object^ Values,
    & System.Object^ FlipAlignment,
    & System.Object^ FlipDimension,
    & System.Object^ LockRotation,
    & System.Object^ Orientation
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ComponentsToCopy
 Array of [components](#) to copy
Repeat

Array of boolean values; each value indicates whether to use the existing mate reference for the corresponding component to copy; if true, copies the existing mate reference; if false, uses the corresponding entry in the NewEntityToMateTo array for the new mate reference

NewEntityToMateTo

Array of new mate references that map to the Repeat array; if an entry in the Repeat array is false, then the corresponding entry in this array is the new entity with which to mate the component to copy

Values

Array of distance or angle values for the mate references; specify distance in meters and angle in radians; valid for distance, angle, and profile center mates only

FlipAlignment

Array of booleans that map to the NewEntityToMateTo array; each value indicates the corresponding mate's alignment; true to flip alignment, false to not

FlipDimension

Array of booleans that map to the Values array; each value indicates the corresponding mate's distance; true for a positive distance dimension, false for a negative distance dimension; valid for distance, angle, and profile center mates only

LockRotation

Array of booleans that map to the NewEntityToMateTo array; true to prevent the components from rotating, false to allow the components to rotate; valid for concentric and profile center mates only

Orientation

Array of longs or integers that map to the Values array; each long or integer indicates the number of clicks in the user interface with which to orient the mate; a positive value indicates to orient the mate clockwise, a negative value indicates to orient the mate counterclockwise; valid for profile center mates only

Return Value

True if calling this method succeeded, false if it failed

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::CopyWithMates2](#).

Example

[Copy Component With Profile Center Mate \(C#\)](#)
[Copy Component With Profile Center Mate \(VB.NET\)](#)
[Copy Component With Profile Center Mate \(VBA\)](#)

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::AddComponent5 Method \(\)](#)
[IAssemblyDoc::AddMate5 Method \(\)](#)

Availability

SOLIDWORKS 2016 FCS, Revision Number 24.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateExplodedView Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : CreateExplodedView Method (IAssemblyDoc)

Creates an explode view of the active assembly configuration.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateExplodedView() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim value As System.Boolean
```

```
value = instance.CreateExplodedView()
```

C#

```
System.bool CreateExplodedView()
```

C++/CLI

```
System.bool CreateExplodedView();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the explode view is successfully created, false if not

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::CreateExplodedView](#).

Example

[Create Exploded Views of an Assembly \(VBA\)](#)
[Create Exploded Views of an Assembly \(VB.NET\)](#)
[Create Exploded Views of an Assembly \(C#\)](#)

Remarks

Call this method multiple times to add multiple explode views to the active assembly configuration.

Perfect explode views often require user interaction. This method attempts to create a perfect explode view based on the current mate conditions.

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::AutoExplode Method \(\)](#)
[IAssemblyDoc::GetExplodedViewConfigurationName Method \(\)](#)
[IAssemblyDoc::GetExplodedViewCount2 Method \(\)](#)
[IAssemblyDoc::GetExplodedViewNames2 Method \(\)](#)
[IAssemblyDoc::ShowExploded2 Method \(\)](#)
[IAssemblyDoc::ViewCollapseAssembly Method \(\)](#)
[IAssemblyDoc::ViewExplodeAssembly Method \(\)](#)
[IModelDocExtension::IsExploded Method \(\)](#)
[IView::IsExploded Method \(\)](#)
[IView::ShowExploded Method \(\)](#)
[IConfiguration::AddRadialExplodeStep Method \(\)](#)
[IConfiguration::AddExplodeStep2 Method \(\)](#)

Availability

SOLIDWORKS 2013 FCS, Revision Number 21.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateMate Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : CreateMate Method (IAssemblyDoc)

Creates a mate with the specified feature data object.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateMate(  
    ByVal MateData As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim MateData As System.Object  
Dim value As System.Object  
  
value = instance.CreateMate(MateData)
```

C#

```
System.object CreateMate(  
    System.object MateData  
)
```

C++/CLI

```
System.Object^ CreateMate(  
& System.Object^ MateData
```

)

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

MateData

Mate-specific object (see **Remarks**)

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::CreateMate](#).

Example

See the Example sections of the interfaces listed in Remarks.

Remarks

[IMateFeatureData](#) is the base interface for the following mate-specific interfaces (an asterisk indicates that pre-selection of mate entities during mate creation is supported):

- [IAngleMateFeatureData](#)
- [ICamFollowerMateFeatureData](#) *
- [ICoincidentMateFeatureData](#) *
- [IConcentricMateFeatureData](#) *
- [IDistanceMateFeatureData](#)
- [IGearMateFeatureData](#) *
- [IHingeMateFeatureData](#) *
- [ILinearCouplerMateFeatureData](#)
- [ILockMateFeatureData](#) *
- [IParallelMateFeatureData](#) *
- [IPerpendicularMateFeatureData](#) *
- [IProfileCenterMateFeatureData](#)
- [IRackPinionMateFeatureData](#) *
- [IScrewMateFeatureData](#) *
- [ISlotMateFeatureData](#) *
- [ISymmetricMateFeatureData](#)
- [ITangentMateFeatureData](#) *
- [IUniversalJointMateFeatureData](#) *
- [IWidthMateFeatureData](#)

To create a standard, advanced, or mechanical mate:

1. Use [IAssemblyDoc::CreateMateData](#) to access a MateFeatureData object.
2. Set the general properties of the MateFeatureData object.
3. Cast the MateFeatureData object to one of the mate-specific objects.
4. Follow specific instructions in the Remarks of the mate-specific interface, set specific properties of the mate-specific object, and/or pre-select entities to mate for asterisked interfaces above.
5. Call this method, passing in the mate-specific object.

To create:

- Misaligned concentric mates, use [IAssemblyDoc::AddConcentricMateWithTolerance](#).
- Cylindrical distance mates, use [IAssemblyDoc::AddDistanceMate](#).

See Also

[IAssemblyDoc Interface](#)

[IAssemblyDoc Members](#)

[IAssemblyDoc::AutoMateRepair Method \(\)](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateMateData Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : CreateMateData Method (IAssemblyDoc)

Creates a mate feature data object for the specified mate type.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateMateData( _  
    ByVal Type As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim Type As System.Integer  
Dim value As System.Object  
  
value = instance.CreateMateData(Type)
```

C#

```
System.object CreateMateData(  
    System.int Type  
)
```

C++/CLI

```
System.Object^ CreateMateData(  
& System.int Type  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Type

Type of mate to create as defined in [swMateType_e](#) (see **Remarks**)

Return Value

[IMateFeatureData](#)

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::CreateMateData](#).

Example

See [IMateFeatureData](#) examples.

Remarks

Type must be one of the following mate types as defined in swMateType_e:

- swMateANGLE
- swMateCAMFOLLOWER
- swMateCOINCIDENT
- swMateCONCENTRIC
- swMateDISTANCE
- swMateGEAR
- swMateHINGE
- swMateLINEARCOUPLER
- swMateLOCK
- swMatePARALLEL
- swMatePERPENDICULAR
- swMatePROFILECENTER
- swMateRACKPINION
- swMateSCREW
- swMateSLOT
- swMateSYMMETRIC
- swMateTANGENT
- swMateUNIVERSALJOINT
- swMateWIDTH

To create a mate, see the [IAssemblydoc::CreateMate](#) Remarks section.

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::AutoMateRepair Method \(\)](#)

Availability

CreateSmartComponent Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : CreateSmartComponent Method (IAssemblyDoc)

Creates a Smart Component.

.NET Syntax

Visual Basic (Declaration)

```

Function CreateSmartComponent( _
    ByVal ComponentIn As Component2, _
    ByVal RelatedComponents As System.Object, _
    ByVal RelatedFeatures As System.Object, _
    ByVal AutoSizeDiameter As System.Boolean, _
    ByVal LpMateReference As Entity, _
    ByVal BoundingValues As System.Object _
) As System.Boolean

```

Visual Basic (Usage)



```

Dim instance As IAssemblyDoc
Dim ComponentIn As Component2
Dim RelatedComponents As System.Object
Dim RelatedFeatures As System.Object
Dim AutoSizeDiameter As System.Boolean
Dim LpMateReference As Entity
Dim BoundingValues As System.Object
Dim value As System.Boolean

value = instance.CreateSmartComponent(ComponentIn, RelatedComponents, RelatedFeatures,
AutoSizeDiameter, LpMateReference, BoundingValues)

```

C#

```

System.bool CreateSmartComponent(
    Component2 ComponentIn,
    System.object RelatedComponents,
    System.object RelatedFeatures,
    System.bool AutoSizeDiameter,
    Entity LpMateReference,
    System.object BoundingValues
)

```

C++/CLI

```

System.bool CreateSmartComponent(
    & Component2^ ComponentIn,
    & System.Object^ RelatedComponents,
    & System.Object^ RelatedFeatures,
    & System.bool AutoSizeDiameter,
    & Entity^ LpMateReference,
    & System.Object^ BoundingValues
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ComponentIn
[IComponent2](#) to make smart

RelatedComponents
 Array of the components to associate with the Smart Component

RelatedFeatures
 Array of the [features](#) contained in the components to associate with the Smart Component

AutoSizeDiameter
 True to auto-size a cylindrical Smart Component that has multiple configurations, false to not

LpMateReference
 Concentric mate reference between a cylindrical face or axis and a feature

BoundingValues
 Array of doubles specifying minimum and maximum diameter values with which each configuration is compatible

Return Value

True if the Smart Component is created, false if not

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::CreateSmartComponent](#).

Example

[Make Smart Component \(VBA\)](#)

[Make Smart Component With Mate \(VBA\)](#)

Remarks

For example, a component with these configurations:

- TenInchDiameter
- ThirteenInchDiameter
- TwentyInchDiameter

might have a BoundingValues array of [9,11,12,14,19,21], which specifies a +1 tolerance over each configuration parameter.

See Also

[IAssemblyDoc Interface](#)

[IAssemblyDoc Members](#)

[IAssemblyDoc::AddSmartComponent Method](#)

[IComponent2::GetSmartComponentData Method](#)

[IComponent2::IsSmartComponent Method](#)

[IComponent2::SetSmartComponentData Method](#)

[ISmartComponentFeatureData Interface](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateSpeedPak Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : CreateSpeedPak Method (IAssemblyDoc)

Creates the specified type of SpeedPak for the active configurations of the selected subassemblies in this assembly.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateSpeedPak( _  
    ByVal SpeedPakType As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim SpeedPakType As System.Integer  
Dim value As System.Boolean
```

```
value = instance.CreateSpeedPak(SpeedPakType)
```

C#

```
System.bool CreateSpeedPak(  
    System.int SpeedPakType  
)
```

C++/CLI

```
System.bool CreateSpeedPak(  
& System.int SpeedPakType  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SpeedPakType

- 1 = Mated
- 2 = Graphics

Return Value

True if SpeedPak is created for the active configurations of the selected subassemblies in this assembly, false if not

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::CreateSpeedPak](#).

Example

[Create SpeedPak for Subassemblies \(C#\)](#)
[Create SpeedPak for Subassemblies \(VB.NET\)](#)
[Create SpeedPak for Subassemblies \(VBA\)](#)

Remarks

If a selected subassembly already has SpeedPak, then calling this method again does not create another SpeedPak, regardless if the type of SpeedPak to create is different than the existing SpeedPak.

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::SetSpeedPakConfigurations Method \(\)](#)
[IAssemblyDoc::SetSpeedPakToParent Method \(\)](#)
[IAssemblyDoc::UpdateSpeedPak Method \(\)](#)
[IAssemblyDoc::UseSpeedPak Method \(\)](#)
[IComponent2::IsSpeedPak Property \(\)](#)
[IConfigurationManager::AddSpeedPak2 Method \(\)](#)

Availability

SOLIDWORKS FCS 2017, Revision Number 25.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

DeleteSelections Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : DeleteSelections Method (IAssemblyDoc)

Delete either the selected components of a subassembly or the subassembly of the selected component.

.NET Syntax

Visual Basic (Declaration)

```
Function DeleteSelections( _  
    ByVal DeleteOptions As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim DeleteOptions As System.Integer  
Dim value As System.Boolean  
  
value = instance.DeleteSelections(DeleteOptions)
```

C#

```
System.bool DeleteSelections(  
    System.int DeleteOptions  
)
```

C++/CLI

```
System.bool DeleteSelections(  
& System.int DeleteOptions  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

DeleteOptions

Type of selection to delete as defined in [swAssemblyDeleteOptions_e](#)

Return Value

True if either the selected components of a subassembly are deleted or the subassembly of the selected component is deleted, false if not

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::DeleteSelections](#).

Example

[Delete Subassemblies \(C#\)](#)
[Delete Subassemblies \(VB.NET\)](#)
[Delete Subassemblies \(VBA\)](#)

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IModelDoc2::EditDelete Method \(\)](#)
[IModelDocExtension::DeleteSelection2 Method \(\)](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

DissolveComponentPattern Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : DissolveComponentPattern Method (IAssemblyDoc)

Dissolves the selected component patterns.

.NET Syntax

Visual Basic (Declaration)

```
Function DissolveComponentPattern() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim value As System.Boolean
```

```
value = instance.DissolveComponentPattern()
```

C#

```
System.bool DissolveComponentPattern()
```

C++/CLI

```
System.bool DissolveComponentPattern();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the selected component patterns are dissolved, false if not

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::DissolveComponentPattern](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12

SOLIDWORKS API Help

[Send comments on this topic.](#)

DissolveSubAssembly Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : DissolveSubAssembly Method (IAssemblyDoc)

Dissolves the selected subassembly in this assembly.

.NET Syntax

Visual Basic (Declaration)

```
Function DissolveSubAssembly() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc
Dim value As System.Boolean

value = instance.DissolveSubAssembly()
```

C#

```
System.bool DissolveSubAssembly()
```

C++/CLI

```
System.bool DissolveSubAssembly();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the subassembly was successfully dissolved, false if not

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::DissolveSubAssembly](#).

Remarks

You must preselect the subassembly to dissolve when you run this method.

This method automatically deletes any features that need to be deleted as a result of the dissolve operation, without input from the user.

This method closes any component files when called in an assembly. If the components were modified, then those modifications are not automatically saved. You must save any modifications before the files are closed.

See Also

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

EditAssembly Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : EditAssembly Method (IAssemblyDoc)

Switches back to the assembly document for editing.

.NET Syntax

Visual Basic (Declaration)

```
Sub EditAssembly()
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
  
instance.EditAssembly()
```

C#

```
void EditAssembly()
```

C++/CLI

```
void EditAssembly();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::EditAssembly](#).

Example

[Create Plane thru 3 Points In-context \(VBA\)](#)
[Suppress Component Feature \(C#\)](#)
[Suppress Component Feature \(VB.NET\)](#)
[Suppress Component Feature \(VBA\)](#)
[Insert Cavity \(C#\)](#)
[Insert Cavity \(VB.NET\)](#)
[Insert Cavity \(VBA\)](#)

Remarks

This method is used with [IAssemblyDoc::EditPart2](#), which allows you to edit a part within the context of an assembly.

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

EditConcentricMate Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : EditConcentricMate Method (IAssemblyDoc)

Edits a misaligned concentric mate.

.NET Syntax

```

Sub EditConcentricMate( _
    ByVal AlignFromEnum As System.Integer, _
    ByVal ConcentricPositionType As System.Integer, _
    ByVal ConcentricToleranceCheck As System.Boolean, _
    ByVal ConcentricToleranceValue As System.Double, _
    ByRef ErrorStatus As System.Integer _
)

```



```

Dim instance As IAssemblyDoc
Dim AlignFromEnum As System.Integer
Dim ConcentricPositionType As System.Integer
Dim ConcentricToleranceCheck As System.Boolean
Dim ConcentricToleranceValue As System.Double
Dim ErrorStatus As System.Integer

instance.EditConcentricMate(AlignFromEnum, ConcentricPositionType, ConcentricToleranceCheck,
ConcentricToleranceValue, ErrorStatus)

```

```

void EditConcentricMate(
    System.int AlignFromEnum,
    System.int ConcentricPositionType,
    System.bool ConcentricToleranceCheck,
    System.double ConcentricToleranceValue,
    out System.int ErrorStatus
)

```

```

void EditConcentricMate(
& System.int AlignFromEnum,
& System.int ConcentricPositionType,
& System.bool ConcentricToleranceCheck,
& System.double ConcentricToleranceValue,
& [Out] System.int ErrorStatus
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AlignFromEnum
Type of mate alignment as defined in [swMateAlign_e](#)

ConcentricPositionType
Misaligned concentric mate position as defined in [swConcentricAlignmentType_e](#)

ConcentricToleranceCheck
True to override the deviation, false to have SOLIDWORKS calculate the deviation

ConcentricToleranceValue
Maximum deviation; valid only when ConcentricToleranceCheck is true

ErrorStatus
Success or error as defined by [swAddMateError_e](#)

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::EditConcentricMate](#).

Example

See the [IAssemblyDoc::AddConcentricMateWithTolerance](#) examples.

Remarks

Select these entities before calling this method:

- Two entities that define the misaligned concentric mate
- Misaligned concentric mate

A parent non-misaligned (i.e., standard) concentric mate must already exist for the selected misaligned concentric mate. If a suitable parent concentric mate does not exist, then the selected misaligned concentric mate changes to a standard concentric mate. You can also change a misaligned concentric mate to a standard concentric mate using [IAssemblyDoc::EditMate4](#).

After calling `IAssemblyDoc::EditConcentricMate`, call [IModelDoc2::EditRebuild3](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::AddConcentricMateWithTolerance Method \(\)](#)
[IAssemblyDoc::AddMate5 Method \(\)](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

EditDistanceMate Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : EditDistanceMate Method (IAssemblyDoc)

Edits a distance mate.

.NET Syntax

Visual Basic (Declaration)

```
Sub EditDistanceMate( _  
    ByVal AlignFromEnum As System.Integer, _  
    ByVal Flip As System.Boolean, _  
    ByVal Distance As System.Double, _  
    ByVal DistanceAbsUpperLimit As System.Double, _  
    ByVal DistanceAbsLowerLimit As System.Double, _  
    ByVal FirstArcCondition As System.Integer, _  
    ByVal SecondArcCondition As System.Integer, _  
    ByRef ErrorStatus As System.Integer _  
)
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim AlignFromEnum As System.Integer  
Dim Flip As System.Boolean  
Dim Distance As System.Double  
Dim DistanceAbsUpperLimit As System.Double  
Dim DistanceAbsLowerLimit As System.Double  
Dim FirstArcCondition As System.Integer  
Dim SecondArcCondition As System.Integer  
Dim ErrorStatus As System.Integer  
  
instance.EditDistanceMate(AlignFromEnum, Flip, Distance, DistanceAbsUpperLimit,  
DistanceAbsLowerLimit, FirstArcCondition, SecondArcCondition, ErrorStatus)
```

C#

```
void EditDistanceMate(  
    System.int AlignFromEnum,  
    System.bool Flip,  
    System.double Distance,  
    System.double DistanceAbsUpperLimit,  
    System.double DistanceAbsLowerLimit,  
    System.int FirstArcCondition,  
    System.int SecondArcCondition,  
    out System.int ErrorStatus  
)
```

C++/CLI

```
void EditDistanceMate(  
& System.int AlignFromEnum,  
& System.bool Flip,
```

```

& System.double Distance,
& System.double DistanceAbsUpperLimit,
& System.double DistanceAbsLowerLimit,
& System.int FirstArcCondition,
& System.int SecondArcCondition,
& [Out] System.int ErrorStatus
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AlignFromEnum

Type of alignment as defined in [swMateAlign_e](#)

Flip

True to flip the mate entities, false to not (see **Remarks**)

Distance

Distance value (see **Remarks**)

DistanceAbsUpperLimit

Absolute maximum distance value (see **Remarks**)

DistanceAbsLowerLimit

Absolute minimum distance value (see **Remarks**)

FirstArcCondition

First arc condition as defined in [swDistanceMateArcConditions_e](#); valid only for cylindrical distance mates (see **Remarks**)

SecondArcCondition

Second arc condition as defined in [swDistanceMateArcConditions_e](#); valid only for cylindrical distance mates (see **Remarks**)

ErrorStatus

Success or error as defined by [swAddMateError_e](#)

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::EditDistanceMate](#).

Example

[Add and Edit Distance Mate \(VBA\)](#)

[Add and Edit Distance Mate \(VB.NET\)](#)

[Add and Edit Distance Mate \(C#\)](#)

Remarks

Select these entities before calling this method:

- Two entities that define the distance mate (use [IEntity::Select4](#) with Data = Nothing)
- Distance mate feature (use [IFeature::Select2](#) with Mark = 0)

If the mate is applied to the closest position that meets the mate condition specified by Distance, then setting Flip to true moves the components to the other possible mate position.

Call [IModelDoc2::EditRebuild3](#) after calling this method.

See [IAssemblyDoc::AddDistanceMate](#) Remarks.

See Also

[IAssemblyDoc Interface](#)

[IAssemblyDoc Members](#)

[IAssemblyDoc::EditMate4 Method \(\)](#)

[IMate2::DistanceFirstArcCondition Property \(\)](#)

[IMate2::DistanceSecondArcCondition Property \(\)](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

EditMate Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : EditMate Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::EditMate2](#).

.NET Syntax

Visual Basic (Declaration)

```

Sub EditMate( _
    ByVal MateType As System.Integer, _
    ByVal Align As System.Integer, _
    ByVal Flip As System.Boolean, _
    ByVal Dist As System.Double, _
    ByVal Angle As System.Double _
)

```

Visual Basic (Usage)



```

Dim instance As IAssemblyDoc
Dim MateType As System.Integer
Dim Align As System.Integer
Dim Flip As System.Boolean
Dim Dist As System.Double
Dim Angle As System.Double

instance.EditMate(MateType, Align, Flip, Dist, Angle)

```

C#

```

void EditMate(
    System.int MateType,
    System.int Align,
    System.bool Flip,
    System.double Dist,
    System.double Angle
)

```

C++/CLI

```

void EditMate(
    & System.int MateType,
    & System.int Align,
    & System.bool Flip,
    & System.double Dist,
    & System.double Angle
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

MateType

Align

Flip

Dist

Angle

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::EditMate](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

EditMate2 Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : EditMate2 Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::EditMate3](#).

.NET Syntax

```

Sub EditMate2( _
    ByVal MateTypeFromEnum As System.Integer, _
    ByVal AlignFromEnum As System.Integer, _
    ByVal Flip As System.Boolean, _
    ByVal Distance As System.Double, _
    ByVal DistanceAbsUpperLimit As System.Double, _
    ByVal DistanceAbsLowerLimit As System.Double, _
    ByVal GearRatioNumerator As System.Double, _
    ByVal GearRatioDenominator As System.Double, _
    ByVal Angle As System.Double, _
    ByVal AngleAbsUpperLimit As System.Double, _
    ByVal AngleAbsLowerLimit As System.Double, _
    ByRef ErrorStatus As System.Integer _
)

```

```

Dim instance As IAssemblyDoc
Dim MateTypeFromEnum As System.Integer
Dim AlignFromEnum As System.Integer
Dim Flip As System.Boolean
Dim Distance As System.Double
Dim DistanceAbsUpperLimit As System.Double
Dim DistanceAbsLowerLimit As System.Double
Dim GearRatioNumerator As System.Double
Dim GearRatioDenominator As System.Double
Dim Angle As System.Double
Dim AngleAbsUpperLimit As System.Double
Dim AngleAbsLowerLimit As System.Double
Dim ErrorStatus As System.Integer

```

```

instance.EditMate2(MateTypeFromEnum, AlignFromEnum, Flip, Distance, DistanceAbsUpperLimit,
DistanceAbsLowerLimit, GearRatioNumerator, GearRatioDenominator, Angle, AngleAbsUpperLimit,
AngleAbsLowerLimit, ErrorStatus)

```

```

void EditMate2(
    System.int MateTypeFromEnum,
    System.int AlignFromEnum,
    System.bool Flip,
    System.double Distance,
    System.double DistanceAbsUpperLimit,
    System.double DistanceAbsLowerLimit,
    System.double GearRatioNumerator,
    System.double GearRatioDenominator,
    System.double Angle,
    System.double AngleAbsUpperLimit,
    System.double AngleAbsLowerLimit,
    out System.int ErrorStatus
)

```

```

void EditMate2(
& System.int MateTypeFromEnum,
& System.int AlignFromEnum,
& System.bool Flip,
& System.double Distance,
& System.double DistanceAbsUpperLimit,
& System.double DistanceAbsLowerLimit,
& System.double GearRatioNumerator,
& System.double GearRatioDenominator,
& System.double Angle,
& System.double AngleAbsUpperLimit,
& System.double AngleAbsLowerLimit,
& [Out] System.int ErrorStatus
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

MateTypeFromEnum

Type of mate as defined in [swMateType_e](#)

AlignFromEnum

Type of alignment desired as defined in [swMateAlign_e](#)

Flip

True to flip the component, false to not

Distance

Distance value to use with distance or limit mates

DistanceAbsUpperLimit

Absolute maximum distance allowed

DistanceAbsLowerLimit

Absolute minimum distance allowed

GearRatioNumerator

Gear ratio numerator value for gear mates

GearRatioDenominator

Gear ratio denominator value for gear mates

Angle

Angle value to use with angle mates

AngleAbsUpperLimit

Absolute maximum angle allowed

AngleAbsLowerLimit

Absolute minimum angle allowed

ErrorStatus

Success or error as defined by [swAddMateError_e](#)

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::EditMate2](#).

Remarks

The first selection should be the two items that are mated (that is, two faces, edge and face, and so on), and the third selection should be the mate feature to be edited. The mate feature must be selected last. The two mated items must be selected with a selection mark of 1. See [ISelectData](#) or [IModelDocExtension::SelectByID2](#) for details on using selection marks.

If mateType is swMateDISTANCE or swMateANGLE when the mate is applied to the closest position that meets the mate condition specified by distance or angle, then setting flip to True moves the assembly to the other possible mate position.

This method does not support editing **Inplace** mates.

See Also

[IAssemblyDoc Interface](#)

[IAssemblyDoc Members](#)

[IAssemblyDoc::AddMate3 Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12

SOLIDWORKS API Help

[Send comments on this topic.](#)

EditMate3 Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : EditMate3 Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::EditMate4](#).

Visual Basic (Declaration)

```
Sub EditMate3( _
    ByVal MateTypeFromEnum As System.Integer, _
    ByVal AlignFromEnum As System.Integer, _
    ByVal Flip As System.Boolean, _
    ByVal Distance As System.Double, _
    ByVal DistanceAbsUpperLimit As System.Double, _
    ByVal DistanceAbsLowerLimit As System.Double, _
    ByVal GearRatioNumerator As System.Double, _
    ByVal GearRatioDenominator As System.Double, _
    ByVal Angle As System.Double, _
    ByVal AngleAbsUpperLimit As System.Double, _
    ByVal AngleAbsLowerLimit As System.Double, _
    ByVal ForPositioningOnly As System.Boolean, _
    ByVal LockRotation As System.Boolean, _
    ByVal WidthMateOption As System.Integer, _
    ByRef ErrorStatus As System.Integer _
)
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc
Dim MateTypeFromEnum As System.Integer
Dim AlignFromEnum As System.Integer
Dim Flip As System.Boolean
Dim Distance As System.Double
Dim DistanceAbsUpperLimit As System.Double
Dim DistanceAbsLowerLimit As System.Double
Dim GearRatioNumerator As System.Double
Dim GearRatioDenominator As System.Double
Dim Angle As System.Double
Dim AngleAbsUpperLimit As System.Double
Dim AngleAbsLowerLimit As System.Double
Dim ForPositioningOnly As System.Boolean
Dim LockRotation As System.Boolean
Dim WidthMateOption As System.Integer
Dim ErrorStatus As System.Integer
```

```
instance.EditMate3(MateTypeFromEnum, AlignFromEnum, Flip, Distance, DistanceAbsUpperLimit,
DistanceAbsLowerLimit, GearRatioNumerator, GearRatioDenominator, Angle, AngleAbsUpperLimit,
AngleAbsLowerLimit, ForPositioningOnly, LockRotation, WidthMateOption, ErrorStatus)
```

C#

```
void EditMate3(
    System.int MateTypeFromEnum,
    System.int AlignFromEnum,
    System.bool Flip,
    System.double Distance,
    System.double DistanceAbsUpperLimit,
    System.double DistanceAbsLowerLimit,
    System.double GearRatioNumerator,
    System.double GearRatioDenominator,
    System.double Angle,
    System.double AngleAbsUpperLimit,
    System.double AngleAbsLowerLimit,
    System.bool ForPositioningOnly,
    System.bool LockRotation,
    System.int WidthMateOption,
    out System.int ErrorStatus
)
```

C++/CLI

```
void EditMate3(
& System.int MateTypeFromEnum,
```

```

& System.int AlignFromEnum,
& System.bool Flip,
& System.double Distance,
& System.double DistanceAbsUpperLimit,
& System.double DistanceAbsLowerLimit,
& System.double GearRatioNumerator,
& System.double GearRatioDenominator,
& System.double Angle,
& System.double AngleAbsUpperLimit,
& System.double AngleAbsLowerLimit,
& System.bool ForPositioningOnly,
& System.bool LockRotation,
& System.int WidthMateOption,
& [Out] System.int ErrorStatus
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

MateTypeFromEnum

Type of mate as defined in [swMateType_e](#)

AlignFromEnum

Type of alignment as defined in [swMateAlign_e](#)

Flip

True to flip the component, false to not

Distance

Distance value; valid only if MateTypeFromEnum is swMateType_e.swMateDISTANCE

DistanceAbsUpperLimit

Absolute maximum distance allowed; valid only if MateTypeFromEnum is swMateType_e.swMateDISTANCE

DistanceAbsLowerLimit

Absolute minimum distance allowed; valid only if MateTypeFromEnum is swMateType_e.swMateDISTANCE

GearRatioNumerator

Gear ratio numerator value; valid only if MateTypeFromEnum is swMateType_e.swMateGEAR

GearRatioDenominator

Gear ratio denominator value; valid only if MateTypeFromEnum is swMateType_e.swMateGEAR

Angle

Angle value; valid only if MateTypeFromEnum is swMateType_e.swMateANGLE

AngleAbsUpperLimit

Absolute maximum angle allowed; valid only if MateTypeFromEnum is swMateType_e.swMateANGLE

AngleAbsLowerLimit

Absolute minimum angle allowed; valid only if MateTypeFromEnum is swMateType_e.swMateANGLE

ForPositioningOnly

True to only position the components according to the mating relationship and not return a mate, false to return a mate

LockRotation

True to lock component rotation, false to not

WidthMateOption

Width mate options as defined in [swMateWidthOptions_e](#); valid only if MateTypeFromEnum is swMateType_e.swMateWIDTH

ErrorStatus

Success or error as defined by [swAddMateError_e](#)

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::EditMate3](#).

Example

[Edit Mate \(VBA\)](#)

[Edit Mate \(VB.NET\)](#)

[Edit Mate \(C#\)](#)

Remarks

The first selections should be the two items that are mated (that is, two faces, edge and face, and so on), and the third selection should be the mate feature to be edited. The mate feature must be selected last. The two mated items must be selected with a selection mark of 1. See [ISelectData](#) or [IModelDocExtension::SelectByID2](#) for details on using selection marks.

If MateTypeFromEnum is swMateType_e.swMateDISTANCE or swMateType_e.swMateANGLE, and the mate is applied to the closest position that meets the mate condition specified by Distance or Angle, then setting Flip to true moves the assembly to the other possible mate position.

This method does not support editing **Inplace** mates.

See Also

[IAssemblyDoc Interface](#)

[IAssemblyDoc Members](#)

[IAssemblyDoc::AddMate5](#)

Availability

EditMate4 Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : EditMate4 Method (IAssemblyDoc)

Obsolete. See the Remarks in each standard, advanced, and mechanical mate's feature data interface.

.NET Syntax

Visual Basic (Declaration)

```

Sub EditMate4( _
    ByVal MateTypeFromEnum As System.Integer, _
    ByVal AlignFromEnum As System.Integer, _
    ByVal Flip As System.Boolean, _
    ByVal Distance As System.Double, _
    ByVal DistanceAbsUpperLimit As System.Double, _
    ByVal DistanceAbsLowerLimit As System.Double, _
    ByVal GearRatioNumerator As System.Double, _
    ByVal GearRatioDenominator As System.Double, _
    ByVal Angle As System.Double, _
    ByVal AngleAbsUpperLimit As System.Double, _
    ByVal AngleAbsLowerLimit As System.Double, _
    ByVal ForPositioningOnly As System.Boolean, _
    ByVal LockRotation As System.Boolean, _
    ByVal WidthMateOption As System.Integer, _
    ByVal RepairMatesWithSameMissingEntity As System.Boolean, _
    ByRef ErrorStatus As System.Integer _
)

```

Visual Basic (Usage)



```

Dim instance As IAssemblyDoc
Dim MateTypeFromEnum As System.Integer
Dim AlignFromEnum As System.Integer
Dim Flip As System.Boolean
Dim Distance As System.Double
Dim DistanceAbsUpperLimit As System.Double
Dim DistanceAbsLowerLimit As System.Double
Dim GearRatioNumerator As System.Double
Dim GearRatioDenominator As System.Double
Dim Angle As System.Double
Dim AngleAbsUpperLimit As System.Double
Dim AngleAbsLowerLimit As System.Double
Dim ForPositioningOnly As System.Boolean
Dim LockRotation As System.Boolean
Dim WidthMateOption As System.Integer
Dim RepairMatesWithSameMissingEntity As System.Boolean
Dim ErrorStatus As System.Integer

```

```

instance.EditMate4(MateTypeFromEnum, AlignFromEnum, Flip, Distance, DistanceAbsUpperLimit,
DistanceAbsLowerLimit, GearRatioNumerator, GearRatioDenominator, Angle, AngleAbsUpperLimit,
AngleAbsLowerLimit, ForPositioningOnly, LockRotation, WidthMateOption,
RepairMatesWithSameMissingEntity, ErrorStatus)

```

C#

```

void EditMate4(
    System.int MateTypeFromEnum,
    System.int AlignFromEnum,
    System.bool Flip,
    System.double Distance,
    System.double DistanceAbsUpperLimit,
    System.double DistanceAbsLowerLimit,
    System.double GearRatioNumerator,
    System.double GearRatioDenominator,
    System.double Angle,

```

```

        System.double AngleAbsUpperLimit,
        System.double AngleAbsLowerLimit,
        System.bool ForPositioningOnly,
        System.bool LockRotation,
        System.int WidthMateOption,
        System.bool RepairMatesWithSameMissingEntity,
        out System.int ErrorStatus
    )

```

C++/CLI

```

void EditMate4(
    & System.int MateTypeFromEnum,
    & System.int AlignFromEnum,
    & System.bool Flip,
    & System.double Distance,
    & System.double DistanceAbsUpperLimit,
    & System.double DistanceAbsLowerLimit,
    & System.double GearRatioNumerator,
    & System.double GearRatioDenominator,
    & System.double Angle,
    & System.double AngleAbsUpperLimit,
    & System.double AngleAbsLowerLimit,
    & System.bool ForPositioningOnly,
    & System.bool LockRotation,
    & System.int WidthMateOption,
    & System.bool RepairMatesWithSameMissingEntity,
    & [Out] System.int ErrorStatus
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

MateTypeFromEnum

Type of mate as defined in [swMateType_e](#)

AlignFromEnum

Type of alignment as defined in [swMateAlign_e](#)

Flip

True to flip the mate components, false to not

Distance

Distance value; valid only if MateTypeFromEnum is swMateType_e.swMateDISTANCE

DistanceAbsUpperLimit

Absolute maximum distance allowed; valid only if MateTypeFromEnum is swMateType_e.swMateDISTANCE

DistanceAbsLowerLimit

Absolute minimum distance allowed; valid only if MateTypeFromEnum is swMateType_e.swMateDISTANCE

GearRatioNumerator

Gear ratio numerator value; valid only if MateTypeFromEnum is swMateType_e.swMateGEAR

GearRatioDenominator

Gear ratio denominator value; valid only if MateTypeFromEnum is swMateType_e.swMateGEAR

Angle

Angle value; valid only if MateTypeFromEnum is swMateType_e.swMateANGLE

AngleAbsUpperLimit

Absolute maximum angle allowed; valid only if MateTypeFromEnum is swMateType_e.swMateANGLE

AngleAbsLowerLimit

Absolute minimum angle allowed; valid only if MateTypeFromEnum is swMateType_e.swMateANGLE

ForPositioningOnly

True to only position the components according to the mating relationship and not return a mate, false to return a mate

LockRotation

True to lock component rotation, false to not

WidthMateOption

Width mate options as defined in [swMateWidthOptions_e](#); valid only if MateTypeFromEnum is swMateType_e.swMateWIDTH

RepairMatesWithSameMissingEntity

True to repair all mates missing the same mate entity, false to not

ErrorStatus

Success or error as defined by [swAddMateError_e](#)

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::EditMate4](#).

Example

[Repair Mates Missing Same Mate Entity \(C#\)](#)
[Repair Mates Missing Same Mate Entity \(VB.NET\)](#)
[Repair Mates Missing Same Mate Entity \(VBA\)](#)

Remarks

To edit:

- Standard, advanced, and mechanical mates created using [IAssemblyDoc::CreateMate](#), follow the steps to edit the mate in the Remarks of the mate's feature data interface.

- Misaligned concentric mates, use [IAssemblyDoc::EditConcentricMate](#).
- Distance mates, use [IAssemblyDoc::EditDistanceMate](#).

If repairing all mates missing the same mate entity, then select the following items in the following order before calling this method:

1. Mate feature to repair.
2. Two model items to mate (that is, two faces, edge and face, and so on). The two model items must be selected with a selection mark of 1. See [ISelectData](#) or [IModelDocExtension::SelectByID2](#) for details on using selection marks.

If `MateTypeFromEnum` is `swMateType_e.swMateDISTANCE` or `swMateType_e.swMateANGLE`, and the mate is applied to the closest position that meets the mate condition specified by Distance or Angle, then setting `Flip` to true moves the components to the other possible mate position

This method does not support editing **Inplace** mates.

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::AddMate5 Method \(\)](#)
[IAssemblyDoc::AddConcentricMateWithTolerance Method \(\)](#)

Availability

SOLIDWORKS 2016 FCS, Revision Number 24.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

EditPart Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : EditPart Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::EditPart2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub EditPart()
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc
instance.EditPart()
```

C#

```
void EditPart()
```

C++/CLI

```
void EditPart();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::EditPart](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

EditPart2 Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : EditPart2 Method (IAssemblyDoc)

Edits the selected part within the context of an assembly.

.NET Syntax

Visual Basic (Declaration)

```
Function EditPart2( _  
    ByVal Silent As System.Boolean, _  
    ByVal AllowReadOnly As System.Boolean, _  
    ByRef Information As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim Silent As System.Boolean  
Dim AllowReadOnly As System.Boolean  
Dim Information As System.Integer  
Dim value As System.Integer  
  
value = instance.EditPart2(Silent, AllowReadOnly, Information)
```

C#

```
System.int EditPart2(  
    System.bool Silent,  
    System.bool AllowReadOnly,  
    ref System.int Information  
)
```

C++/CLI

```
System.int EditPart2(  
& System.bool Silent,  
& System.bool AllowReadOnly,  
& System.int% Information  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Silent

True to suppress messages to user, false to not

AllowReadOnly

True to allow editing of read-only parts, false to not

Information

Status as defined in [swEditPartCommandStatus_e](#)

Return Value

swEditPartSuccessful if successful

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::EditPart2](#).

Example

[Create Plane thru 3 Points In-context \(VBA\)](#)
[Suppress Component Feature \(C#\)](#)
[Suppress Component Feature \(VB.NET\)](#)
[Suppress Component Feature \(VBA\)](#)
[Insert Cavity \(C#\)](#)
[Insert Cavity \(VB.NET\)](#)
[Insert Cavity \(VBA\)](#)

Remarks

This method allows you to control the display of dialog boxes and edit a read-only document. To return to editing the assembly, use [IAssemblyDoc::EditAssembly](#).

See Also

Availability

SOLIDWORKS 2000 FCS, Revision Number 8.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

EditRebuild Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : EditRebuild Method (IAssemblyDoc)

Obsolete. Superseded by [IModelDoc2::EditRebuild3](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub EditRebuild()
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
instance.EditRebuild()
```

C#

```
void EditRebuild()
```

C++/CLI

```
void EditRebuild();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::EditRebuild](#).

See Also

SOLIDWORKS API Help

[Send comments](#) on this topic.

EnableAssemblyRebuild Property (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : EnableAssemblyRebuild Property (IAssemblyDoc)

Gets or sets whether to suspend the rebuild of the assembly.

.NET Syntax

Visual Basic (Declaration)

```
Property EnableAssemblyRebuild As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim value As System.Boolean  
  
instance.EnableAssemblyRebuild = value  
value = instance.EnableAssemblyRebuild
```

C#

```
System.bool EnableAssemblyRebuild {get; set;}
```

C++/CLI

```
property System.bool EnableAssemblyRebuild {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to suspend rebuilding the assembly, false to rebuild the assembly (false is the default)

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::EnableAssemblyRebuild](#).

Example

[Suspend Automatic Rebuilds \(VBA\)](#)

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13

SOLIDWORKS API Help

[Send comments on this topic.](#)

EnablePresentation Property (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : EnablePresentation Property (IAssemblyDoc)

Gets or sets whether the assembly is in presentation mode.

.NET Syntax

Visual Basic (Declaration)

```
Property EnablePresentation As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim value As System.Boolean
```

```
instance.EnablePresentation = value
```

```
value = instance.EnablePresentation
```

C#

```
System.bool EnablePresentation {get; set;}
```

C++/CLI

```
property System.bool EnablePresentation {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True enables presentation mode, false disables it

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::EnablePresentation](#).

Remarks

Call this property before calling [IComponent2::PresentationTransform](#) because presentation transforms are effective only in this mode. This property prohibits reconfiguring the FeatureManager design tree when SOLIDWORKS is using the presentation transforms. This property also prohibits selections in the model view.

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IComponent2::GetTotalTransform Method](#)
[IComponent2::RemovePresentationTransform Method](#)
[IComponent2::Transform2 Property](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ExitIsolate Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : ExitIsolate Method (IAssemblyDoc)

Exits [isolating the selected components](#) and returns the assembly to its original display state.

.NET Syntax

Visual Basic (Declaration)

```
Function ExitIsolate() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim value As System.Boolean  
  
value = instance.ExitIsolate()
```

C#

```
System.bool ExitIsolate()
```

C++/CLI

```
System.bool ExitIsolate();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if isolating the selected component exits and the assembly returns to its original display state, false if not

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::ExitIsolate](#).

Example

[Isolate Component \(C#\)](#)
[Isolate Component \(VB.NET\)](#)
[Isolate Component \(VBA\)](#)

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::SaveIsolate Method \(\)](#)
[IAssemblyDoc::SetIsolateVisibility Method \(\)](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

FeatureByName Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : FeatureByName Method (IAssemblyDoc)

Returns the [IFeature](#) object for the named feature in the assembly.

.NET Syntax

Visual Basic (Declaration)

```
Function FeatureByName(  
    ByVal Name As System.String _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim Name As System.String  
Dim value As System.Object  
  
value = instance.FeatureByName(Name)
```

C#

```
System.object FeatureByName(  
    System.string Name  
)
```

C++/CLI

```
System.Object^ FeatureByName(  
& System.String^ Name  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Name

Name of the feature

Return Value

Object for the feature

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::FeatureByName](#).

Example

[Get Parent-Child Relationship for Component \(VBA\)](#)
[Create and Name Plane \(VBA\)](#)
[Move Assembly Components to New Folder \(C#\)](#)
[Move Assembly Components to New Folder \(VB.NET\)](#)
[Move Assembly Components to New Folder \(VBA\)](#)
[Roll Back Model \(C#\)](#)
[Roll Back Model \(VB.NET\)](#)
[Roll Back Model \(VBA\)](#)
[Suppress Component Feature \(C#\)](#)
[Suppress Component Feature \(VB.NET\)](#)
[Suppress Component Feature \(VBA\)](#)

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::IFeatureByName Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

FeatureExtrusion Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : FeatureExtrusion Method (IAssemblyDoc)

Obsolete. Superseded by [IFeatureManager::FeatureExtrusion](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub FeatureExtrusion()
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
instance.FeatureExtrusion()
```

C#

```
void FeatureExtrusion()
```

C++/CLI

```
void FeatureExtrusion();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::FeatureExtrusion](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

FileDeriveComponentPart Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : FileDeriveComponentPart Method (IAssemblyDoc)

Creates a new part document from the currently selected assembly component.

.NET Syntax

Visual Basic (Declaration)

```
Sub FileDeriveComponentPart()
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
instance.FileDeriveComponentPart()
```

C#

```
void FileDeriveComponentPart()
```

C++/CLI

```
void FileDeriveComponentPart();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::FileDeriveComponentPart](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

FixComponent Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : FixComponent Method (IAssemblyDoc)

Fixes the selected component; i.e., makes it immovable.

.NET Syntax

Visual Basic (Declaration)

```
Sub FixComponent()
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc
```

```
instance.FixComponent()
```

C#

```
void FixComponent()
```

C++/CLI

```
void FixComponent();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::FixComponent](#).

Example

[Resolve All Components and Fix a Component \(C#\)](#)
[Resolve All Components and Fix a Component \(VB.NET\)](#)
[Resolve All Components and Fix a Component \(VBA\)](#)

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::UnfixComponent Method \(\)](#)
[IAssemblyDoc::TemporaryFixGroup Method \(\)](#)
[IAssemblyDoc::TemporaryFixGroupExit Method \(\)](#)
[IComponent2::IsFixed Method \(\)](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

ForceRebuild Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : ForceRebuild Method (IAssemblyDoc)

Obsolete. Superseded by [IModelDoc2::ForceRebuild3](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub ForceRebuild()
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
  
instance.ForceRebuild()
```

C#

```
void ForceRebuild()
```

C++/CLI

```
void ForceRebuild();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::ForceRebuild](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

ForceRebuild2 Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : ForceRebuild2 Method (IAssemblyDoc)

Obsolete. Superseded by [IModelDoc2::ForceRebuild3](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub ForceRebuild2( _  
    ByVal TopOnly As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim TopOnly As System.Boolean  
  
instance.ForceRebuild2(TopOnly)
```

C#

```
void ForceRebuild2(  
    System.bool TopOnly  
)
```

C++/CLI

```
void ForceRebuild2(  
& System.bool TopOnly  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::ForceRebuild2](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

ForceUpdateElectricalData Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : ForceUpdateElectricalData Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::ForceUpdateElectricalData2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub ForceUpdateElectricalData( _
    ByVal Stream As System.Integer _
)
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc
Dim Stream As System.Integer

instance.ForceUpdateElectricalData(Stream)
```

C#

```
void ForceUpdateElectricalData(
    System.int Stream
)
```

C++/CLI

```
void ForceUpdateElectricalData(
    & System.int Stream
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Stream

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::ForceUpdateElectricalData](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

ForceUpdateElectricalData2 Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : ForceUpdateElectricalData2 Method (IAssemblyDoc)

Forces an update of electrical data.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function ForceUpdateElectricalData2( _  
    ByVal Stream As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim Stream As System.Integer  
Dim value As System.Integer  
  
value = instance.ForceUpdateElectricalData2(Stream)
```

C#

```
System.int ForceUpdateElectricalData2(  
    System.int Stream  
)
```

C++/CLI

```
System.int ForceUpdateElectricalData2(  
&    System.int Stream  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Stream
Stream as defined in swElectricalStreamType_e

Return Value

Status of update as defined in swForceUpdateElectricalDataError_e

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::ForceUpdateElectricalData2](#).

Remarks

Third-party applications should use this method to tell the SOLIDWORKS Design software that they have changed the electrical data. The SOLIDWORKS Design software then re-reads the data to get the updates.

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetActiveGroundPlane Method (IAssemblyDoc)

See Also [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : GetActiveGroundPlane Method (IAssemblyDoc)

Gets the active ground plane for the specified configurations.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function GetActiveGroundPlane( _
    ByVal Config\_opt As System.Integer, _
    ByVal Config\_names As System.Object _
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc
Dim Config_opt As System.Integer
Dim Config_names As System.Object
Dim value As System.Object

value = instance.GetActiveGroundPlane(Config_opt, Config_names)
```

C#

```
System.object GetActiveGroundPlane(
    System.int Config\_opt,
    System.object Config\_names
)
```

C++/CLI

```
System.Object^ GetActiveGroundPlane(
    & System.int Config\_opt,
    & System.Object^ Config\_names
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Config_opt

Configurations from which to retrieve active ground planes as defined in [swInConfigurationOpts_e](#):

- [swThisConfiguration](#)
- [swAllConfigurations](#)
- [swSpecifyConfiguration](#)

(see **Remarks**)

Config_names

Array of the names of configurations from which to retrieve active ground planes; valid only if *Config_opt* is set to [swInConfiguration_e.swSpecifyConfiguration](#) (see **Remarks**)

Return Value

Array of ground plane [features](#)

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::GetActiveGroundPlane](#).

Example

[Insert and Activate Ground Plane \(VBA\)](#)

Remarks

If <i>Config_opt</i> is set to swInConfiguration_e...	Then the returned array contains one active ground plane or null for...
swAllConfigurations	Each configuration in the assembly.
swSpecifyConfiguration	Each configuration in <i>Config_names</i> .
swThisConfiguration	The current configuration.

To populate *config_names*, use [IModelDoc2::GetConfigurationNames](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::ActivateGroundPlane Method \(\)](#)
[IFeatureManager::InsertGroundPlane Method \(\)](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetAdvancedSelection Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : GetAdvancedSelection Method (IAssemblyDoc)

Gets the advanced component selection criteria object for this assembly.

.NET Syntax

Visual Basic (Declaration)

Function GetAdvancedSelection() As AdvancedSelectionCriteria

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim value As AdvancedSelectionCriteria  
  
value = instance.GetAdvancedSelection()
```

C#

```
AdvancedSelectionCriteria GetAdvancedSelection()
```

C++/CLI

```
AdvancedSelectionCriteria^ GetAdvancedSelection();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[IAdvancedSelectionCriteria](#)

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::GetAdvancedSelection](#).

Example

See the [IAdvancedSelectionCriteria](#) examples.

Remarks

For more information, see the **Assemblies > Basic Component Operations > Selecting Components > Advanced Component Selection** topic in the SOLIDWORKS Design user-interface help.

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetBox Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : GetBox Method (IAssemblyDoc)

Gets the bounding box.

■ .NET Syntax

Visual Basic (Declaration)

```
Function GetBox( _  
    ByVal Options As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim Options As System.Integer  
Dim value As System.Object  
  
value = instance.GetBox(Options)
```

C#

```
System.object GetBox(  
    System.int Options  
)
```

C++/CLI

```
System.Object^ GetBox(  
& System.int Options  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Options

Type of bounding box as defined by [swBoundingBoxOptions_e](#)

Return Value

Object containing the two diagonal points that bound the component; the array contains 6 doubles (see **Remarks**)

■ Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::GetBox](#).

■ Example

[Get Assembly Bounding Box Using Assembly \(C#\)](#)
[Get Assembly Bounding Box Using Assembly \(VB.NET\)](#)
[Get Assembly Bounding Box Using Assembly \(VBA\)](#)
[Recalculate Bounding Box \(C#\)](#)
[Recalculate Bounding Box \(VB.NET\)](#)
[Recalculate Bounding Box \(VBA\)](#)

■ Remarks

IMPORTANT: The values returned are approximate and should not be used for comparison or calculation purposes. Furthermore, the bounding box may vary after rebuilding the model. The resulting box encloses the object, but it might not be the tightest box.

The X, Y, Z points returned by SOLIDWORKS are the lower- and upper-diagonal corners that bound the component with the box sides parallel to the X, Y and Z axes. SOLIDWORKS returns box dimensions that enclose the object and are typically close to the minimum possible size.

The return value is an array of doubles as follows:

[XCorner1, YCorner1, ZCorner1, XCorner2, YCorner2, ZCorner2]

■ See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IModelDocExtension::GetVisibleBox Method](#)

[IModelDocExtension::RemoveVisibleBox Method](#)
[IModelDocExtension::SetVisibleBox Method](#)
[IAssemblyDoc::IGetBox Method](#)
[IAssemblyDoc::UpdateBox Method](#)
[IBody2::GetBodyBox Method](#)
[IBody2::IGetBodyBox Method](#)
[IComponent2::GetBox Method](#)
[IComponent2::IGetBox Method](#)
[IFace2::GetBox Method](#)
[IFace2::IGetBox Method](#)
[IFeature::GetBox Method](#)
[IFeature::IGetBox Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetComponentByID Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : GetComponentByID Method (IAssemblyDoc)

Gets a top-level assembly component using its component ID.

.NET Syntax

Visual Basic (Declaration)

```
Function GetComponentByID( _  
    ByVal ID As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim ID As System.Integer  
Dim value As System.Object  
  
value = instance.GetComponentByID(ID)
```

C#

```
System.object GetComponentByID(  
    System.int ID  
)
```

C++/CLI

```
System.Object^ GetComponentByID(  
& System.int ID  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ID

Component ID of top-level assembly component (see **Remarks**)

Return Value

Top-level [component](#)

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::GetComponentByID](#).

Example

[Get Top-level Components Using Component IDs \(C#\)](#)
[Get Top-level Components Using Component IDs \(VB.NET\)](#)
[Get Top-level Components Using Component IDs \(VBA\)](#)

Remarks

Call [IComponent2::GetID](#) before calling this method to get the component ID of the top-level assembly component to pass to this method.

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::GetComponentByName Method \(\)](#)
[IAssemblyDoc::GetComponents Method \(\)](#)
[IAssemblyDoc::IGetComponents Method \(\)](#)

Availability

SOLIDWORKS 2016 FCS, Revision Number 24.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetComponentByName Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : GetComponentByName Method (IAssemblyDoc)

Gets the specified top-level assembly component.

.NET Syntax

Visual Basic (Declaration)

```
Function GetComponentByName( _  
    ByVal CompName As System.String _  
) As Component2
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim CompName As System.String  
Dim value As Component2  
  
value = instance.GetComponentByName(CompName)
```

C#

```
Component2 GetComponentByName(  
    System.string CompName  
)
```

C++/CLI

```
Component2^ GetComponentByName(  
& System.String^ CompName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CompName
Name of the top-level assembly component to get

Return Value

[Component](#)

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::GetComponentByName](#).

Example

[Get Component by Name \(C#\)](#)
[Get Component by Name \(VB.NET\)](#)

[Get Component by Name \(VBA\)](#)

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::GetComponents Method](#)
[IAssemblyDoc::IGetComponents Method](#)
[IAssemblyDoc::GetUnloadedComponentNames Method](#)
[IComponent2::GetSelectByIDString Method](#)
[IComponent2::Name2 Property](#)
[IAssemblyDoc::GetComponentByID Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetComponentCount Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : GetComponentCount Method (IAssemblyDoc)

Gets the number of components in the active configuration of this assembly.

.NET Syntax

Visual Basic (Declaration)

```
Function GetComponentCount( _  
    ByVal ToplevelOnly As System.Boolean _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim ToplevelOnly As System.Boolean  
Dim value As System.Integer  
  
value = instance.GetComponentCount(ToplevelOnly)
```

C#

```
System.int GetComponentCount(  
    System.bool ToplevelOnly  
)
```

C++/CLI

```
System.int GetComponentCount(  
&    System.bool ToplevelOnly  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ToplevelOnly

True to get only the number of components at the top level of the FeatureManager design tree, false to get the number of top level and all child components in the FeatureManager design tree

Return Value

Number of components

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::GetComponentCount](#).

Remarks

Use this method before using [IAssemblyDoc::IGetComponents](#) to determine the size of the array to pass into that method.

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IComponent2::GetChildren Method](#)
[IComponent2::GetParent Method](#)
[IConfiguration::GetRootComponent3 Method](#)
[IConfiguration::IGetRootComponent2 Method](#)
[IConfigurationManager::AddConfiguration Method](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetComponents Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : GetComponents Method (IAssemblyDoc)

Gets all of the components in the active configuration of this assembly.

.NET Syntax

Visual Basic (Declaration)

```
Function GetComponents( _  
    ByVal ToplevelOnly As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim ToplevelOnly As System.Boolean  
Dim value As System.Object  
  
value = instance.GetComponents(ToplevelOnly)
```

C#

```
System.object GetComponents(  
    System.bool ToplevelOnly  
)
```

C++/CLI

```
System.Object^ GetComponents(  
&    System.bool ToplevelOnly  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ToplevelOnly

True to get only the top-level components of the FeatureManager design tree, false to get both the top-level and child components in the FeatureManager design tree

Return Value

Array of [IComponent2](#)s (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::GetComponents](#).

Example

[Insert MidSurface in Components \(C#\)](#)
[Insert MidSurface in Component \(VB.NET\)](#)
[Insert MidSurface in Component \(VBA\)](#)
[Get Mates \(VB.NET\)](#)
[Get Mates \(C#\)](#)

Remarks

The components returned by this method can be in any order. You should not rely on the order to indicate anything about children or parents. If the hierarchy and order are important, then use [IModelDoc2::FirstFeature](#), [IFeature::GetTypeName2](#), [IFeature::GetSpecificFeature2](#), [IComponent2::FirstFeature](#), and [IFeature::GetNextFeature](#) to retrieve your information.

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::IGetComponents Method](#)
[IAssemblyDoc::GetComponentCount Method](#)
[IComponent2::GetChildren Method](#)
[IComponent2::GetParent Method](#)
[IConfiguration::GetRootComponent3 Method](#)
[IConfiguration::IGetRootComponent2 Method](#)
[IConfigurationManager::ActiveConfiguration Property](#)
[IAssemblyDoc::SelectiveOpen Method](#)
[IAssemblyDoc::GetComponentByID Method](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetDragOperator Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : GetDragOperator Method (IAssemblyDoc)

Gets the drag operator for dynamic drag operations in this assembly.

.NET Syntax

Visual Basic (Declaration)

```
Function GetDragOperator() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim value As System.Object  
  
value = instance.GetDragOperator()
```

C#

```
System.object GetDragOperator()
```

C++/CLI

```
System.Object^ GetDragOperator();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Object for the drag operator

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::GetDragOperator](#).

Example

[Rotate Assembly Component on Axis Using IDragOperator::Drag \(C#\)](#)
[Rotate Assembly Component on Axis Using IDragOperator::Drag \(VB.NET\)](#)
[Rotate Assembly Component on Axis Using IDragOperator::Drag \(VBA\)](#)

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::IGetDragOperator Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetDroppedAtEntity Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : GetDroppedAtEntity Method (IAssemblyDoc)

Gets a pointer to the entity where a file is dropped into this assembly.

.NET Syntax

Visual Basic (Declaration)

```
Function GetDroppedAtEntity() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim value As System.Object  
  
value = instance.GetDroppedAtEntity()
```

C#

```
System.object GetDroppedAtEntity()
```

C++/CLI

```
System.Object^ GetDroppedAtEntity();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Object for the entity

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::GetDroppedAtEntity](#).

Remarks

Use this method in the handler for the [FileDropNotify](#) event.

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::SetDroppedFileName Method](#)
[SetDroppedFileName Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetEditTarget Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : GetEditTarget Method (IAssemblyDoc)

Gets the model document that is currently being edited.

.NET Syntax

Visual Basic (Declaration)

```
Function GetEditTarget() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim value As System.Object  
  
value = instance.GetEditTarget()
```

C#

```
System.object GetEditTarget()
```

C++/CLI

```
System.Object^ GetEditTarget();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Object for the model document being edited

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::GetEditTarget](#).

Example

[Suppress Component Feature \(C#\)](#)
[Suppress Component Feature \(VB.NET\)](#)
[Suppress Component Feature \(VBA\)](#)

Remarks

This method is useful when the user is performing an in-context edit of a part component because the currently active document returned from [ISldWorks::ActiveDoc](#) might not be the document that the user is editing. For example, the user can use in-context editing to modify an assembly component. The active document is the assembly, but this method returns the IModelDoc2 for the component being edited.

The IModelDoc2 object returned by this method might be that of the assembly if no part is being in-place edited. In other words, this method returns a valid pointer (non-NULL) even when user is not editing a part in-context.

You can use the IModelDoc2 object returned by this method to determine which assembly component is being edited in-context. In general, you should not use this IModelDoc2 object for feature creation within the component part. Feature creation typically requires the IModelDoc2 of the active document. During feature creation, SOLIDWORKS automatically determines whether the feature should be created and owned by the active assembly, or if it is an in-context edit in which the feature should be created and owned by the component part.

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::GetEditTargetComponent Method](#)
[IGetEditTarget2 Method](#)
[ISlnEditTarget2 Method](#)
[ActiveDoc Property](#)
[IsEditingSelf Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetEditTargetComponent Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : GetEditTargetComponent Method (IAssemblyDoc)

Gets the component that is currently being edited.

.NET Syntax

Visual Basic (Declaration)

Function GetEditTargetComponent() As Component2

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim value As Component2  
  
value = instance.GetEditTargetComponent()
```

C#

```
Component2 GetEditTargetComponent()
```

C++/CLI

```
Component2^ GetEditTargetComponent();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[IComponent2](#) being edited

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::GetEditTargetComponent](#).

Example

[Reorganize Components \(VBA\)](#)

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::GetEditTarget Method](#)
[IGetEditTarget Method](#)
[IsEditingSelf Method](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetExplodedViewConfigurationName Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : GetExplodedViewConfigurationName Method (IAssemblyDoc)

Gets the name of the configuration for the specified exploded view.

.NET Syntax

Visual Basic (Declaration)

```
Function GetExplodedViewConfigurationName( _  
    ByVal ExplodedViewName As System.String _  
) As System.String
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim ExplodedViewName As System.String  
Dim value As System.String  
  
value = instance.GetExplodedViewConfigurationName(ExplodedViewName)
```

C#

```
System.string GetExplodedViewConfigurationName(  
    System.string ExplodedViewName  
)
```

C++/CLI

```
System.String^ GetExplodedViewConfigurationName(  
& System.String^ ExplodedViewName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ExplodedViewName
Name of the exploded view

Return Value

Name of the configuration for the specified exploded view

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::GetExplodedViewConfigurationName](#).

Example

[Get Exploded Views for Configuration \(C#\)](#)
[Get Exploded Views for Configuration \(VB.NET\)](#)
[Get Exploded Views for Configuration \(VBA\)](#)

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::GetExplodedViewCount2 Method \(\)](#)
[IAssemblyDoc::GetExplodedViewNames2 Method \(\)](#)
[IAssemblyDoc::CreateExplodedView Method \(\)](#)
[IAssemblyDoc::AutoExplode Method \(\)](#)
[IAssemblyDoc::ShowExploded2 Method \(\)](#)
[IAssemblyDoc::ViewExplodeAssembly Method \(\)](#)
[IAssemblyDoc::ViewCollapseAssembly Method \(\)](#)
[IModelDocExtension::IsExploded Method \(\)](#)
[IView::IsExploded Method \(\)](#)
[IView::ShowExploded Method \(\)](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0